SOLAR DECODING : Reading The Rays

I. INTRODUCTIONS AND MISSION OBJECTIVES

Voyage-123, an Alliance probe, has collected solar radiation and environmental data from a remote planet. This project analyzes the recorded data using Python to identify trends, relationships, and unusual observations. It also examines daily solar activity and explores the possibility of predicting solar radiation to support the Rebel Alliance's energy planning.

- DATA VISUALIZATION
- DRAWING INFERENCES
- SOLAR DAY ANALYSIS
- SOLAR RADIATION PREDICTION

II. DATASET

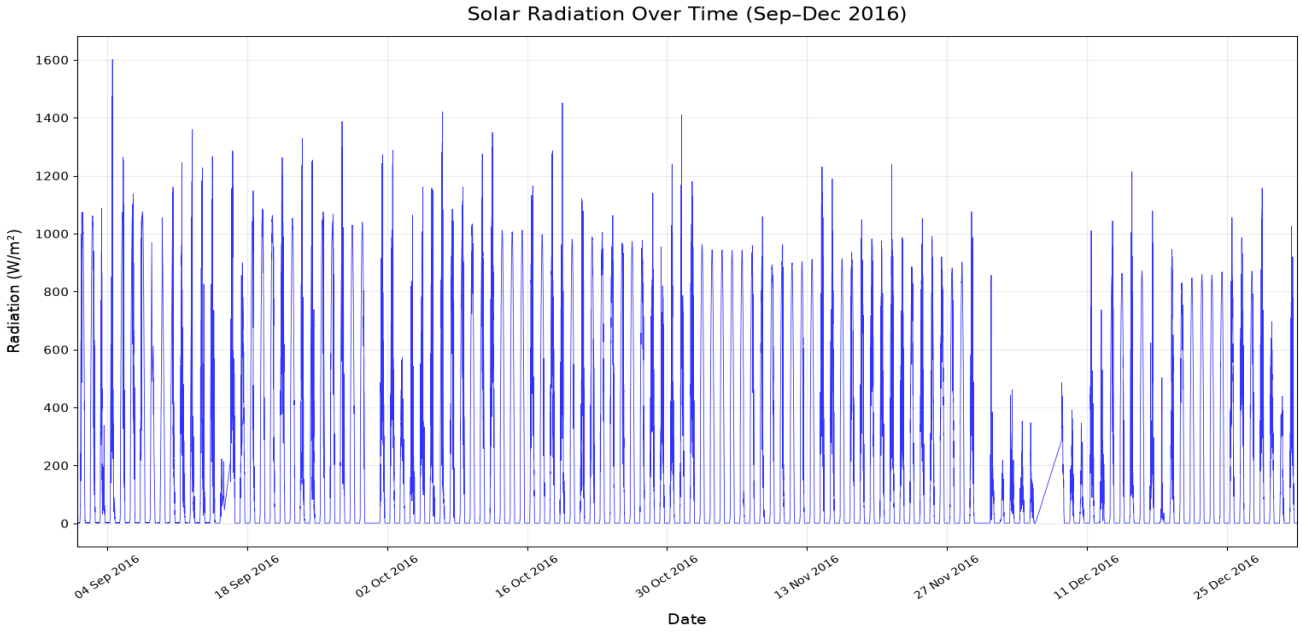
It contains various graph files and solar analysis files consisting of SolarPrediction.csv, imodel.py, SolarDayAnalysis.py ,daily_report.txt and readme.md

III. CONTENT

A. DATA VISUALIZATION:

i. Solar Radiation vs Time:

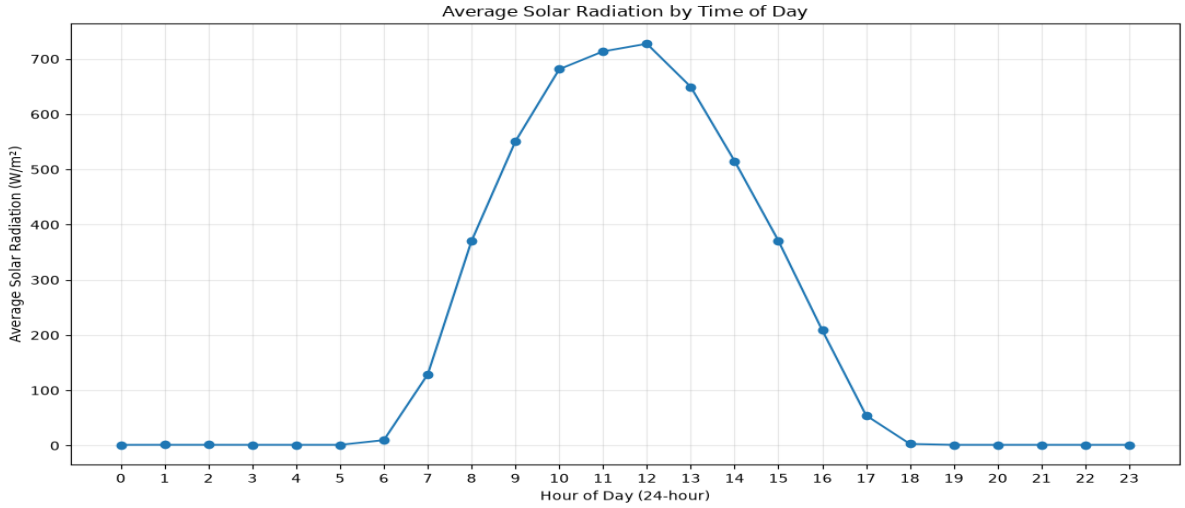
The graph shows how solar radiation changed from September to December 2016



recorded by Voyage 123. The radiation level keeps changing throughout this period. It is generally higher during the daytime and lower at night . There are also some sudden increases and decreases in the readings. This graph helps us understand how solar radiation changes with time.(It can be understood later by the average radiation vs Hour of a Day Graph).

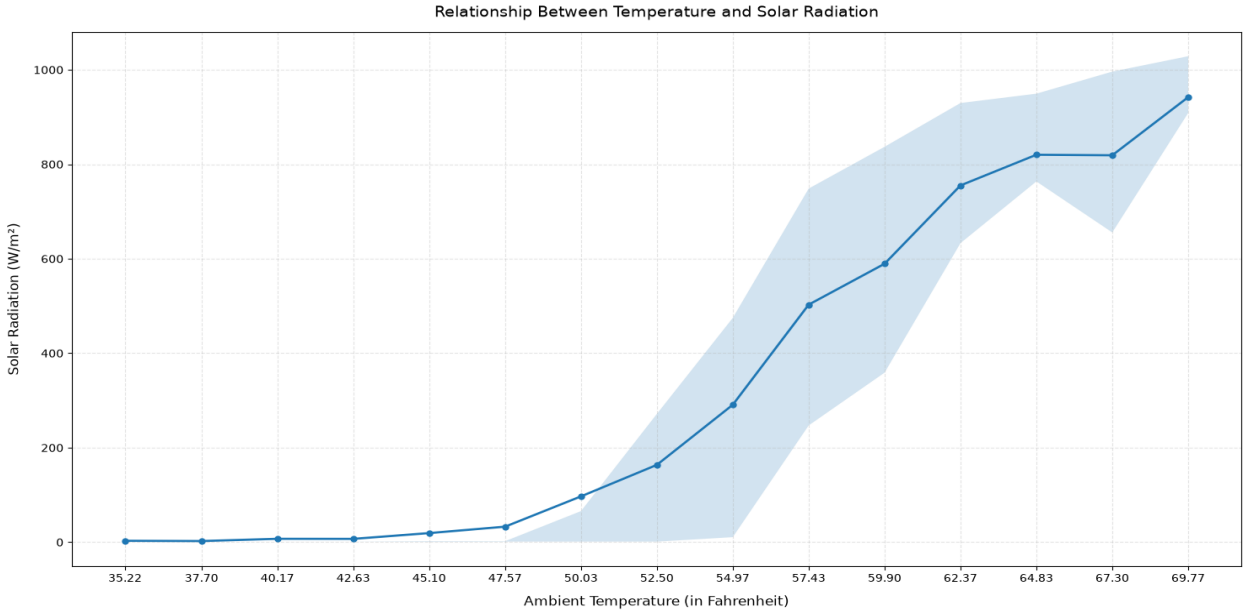
ii. Average Solar Radiation vs Hour of the Day

The graph shows the average solar radiation recorded during each hour of the day. Solar radiation is generally low at night and increases during the morning. It reaches higher levels around midday and then decreases towards the evening. This shows how solar radiation changes throughout the day.



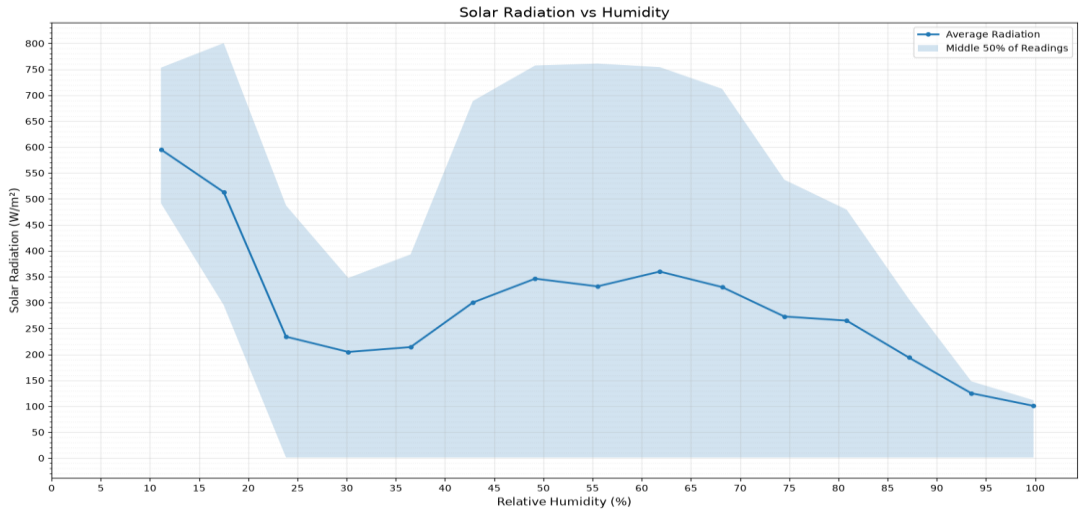
iii. Solar Radiation vs Temperature:

The graph shows the relationship between temperature and solar radiation. We can see that solar radiation generally increases as the temperature rises. This means that higher temperatures are usually linked with higher solar radiation in our dataset. However, the readings do not follow the same pattern in every case, as other factors may also affect solar radiation.



The graph shows how average solar radiation changes across different temperature ranges. We can see that radiation generally increases as temperature rises. The blue shaded area shows the range between the lower and upper quartiles of radiation in each temperature group (interval of 15) . This indicates that radiation values also vary within the same temperature range.

iv. Solar Radiation vs Humidity:



The graph shows how solar radiation changes with different humidity levels. In general, solar radiation tends to decrease as humidity increases, although the pattern is not the same at every level. The blue shaded area shows the middle 50% of the radiation readings in each humidity group. This helps us understand how radiation values vary at different humidity levels.

B. DRAWING INFERENCES:

- Solar radiation is generally highest around midday (around 12), when sunlight is more brighter. It increases during the morning, reaches its peak around noon, and then decreases towards the evening. This shows that the time of day has an important effect on solar radiation (it can also be see through graph ii).
- Temperature shows a strong positive relationship with solar radiation. As temperature increases, solar radiation generally increases as well. Among the parameters analyzed, temperature appears to have the strongest relationship with radiation.
- The graphs show that solar radiation generally increases as temperature rises, indicating a positive relationship. In contrast, solar radiation tends to decrease as humidity increases, showing a negative relationship .It is observed that average humidity generally decreases as temperature increases.
- On 15 September , solar radiation was much lower than than its nearby dates that is 14 and 16 September. The dataset also shows that on this day humidity is very high and temperature is comparatively lower. These conditions may contribute to the reduced solar radiation.

C. SOLAR DAY ANALYSIS:

- Date and time are converted into DateTime columns to smoothly calculate things
- Daily average and daily maximum are calculated and grouped with date then a separate daily_report.txt file is created to store this report
- Solar Radiation is highest on : 4/09/2016 at 12:15:04. Highest value is 1601.26W/m²
- Corresponding terms for this day are:

    - Temperature : 61
    - Humidity : 93
    - Wind Speed : 9.0

- Sun related data for this day:

    - Sunrise time: 06:08:00
    - Sunset time: 18:35:00
    - Time after Sunrise to reach peak radiation: 06:07:04

- Top5 days with maximum average solar radiation:

    - Top 5 days by average radiation:
    - Date Solar Radiation
    - 2016-09-19 335.504400
    - 2016-09-02 333.471219
    - 2016-09-22 325.515880
    - 2016-09-25 319.921514
    - 2016-09-28 312.400390

D. SOLAR RADIATION PREDICTION:

- To build the model LinearRegression library is used in scikit-learn
- All the important features are used to make the model like:

    - Temperature
    - Humidity
    - Pressure
    - Wind speed
    - Wind direction
    - Hour of the day

- The model is trained by the data of SolarPredictions.csv by Voyage-123 to predict the Solar Radiation using the given inputs

- When evaluated and tested it gave the following scores:

  - **RMSE:** 205.9964
  - **R^2 :** 0.5808

- According to the imodel order of importance is:

  1. Pressure
  2. Temperature
  3. Speed
  4. Hour
  5. Wind Direction
  6. Humidity

- Additionally I have also added a system to take input from the user to predict the Solar Radiation by given parameters and added a warning to pop up if the predicted Radiation is larger than 800W/m^2